

SUMMER EXAMINATION 2026
Natural Language Processing - Version A
BSIT - 3rd Semester Time: 90 Minutes Max Marks: 80

Name: _____ Roll No.: _____ Date: _____

1. LSTMs address vanishing gradients using
(A) gates and cell state (B) one-hot vectors
(C) stop words (D) IDF
2. Sentiment analysis classifies
(A) opinion polarity (B) syntax trees only
(C) document length (D) speaker identity only
3. Add-one smoothing adds
(A) one to counts (B) one token to every sentence
(C) one class (D) one hidden layer
4. GRUs use update and reset
(A) gates (B) parse trees
(C) document terms (D) speech codecs only
5. TF-IDF is high for terms frequent in a document but
(A) rare in the corpus (B) common everywhere
(C) absent from the document (D) only punctuation
6. The value vectors in attention carry
(A) content to aggregate (B) class names
(C) grammar rules (D) document IDs only
7. Lowercasing can reduce vocabulary but may lose
(A) case information (B) token order
(C) all punctuation (D) sentence boundaries only
8. The Viterbi algorithm finds the most likely
(A) state sequence (B) document ranking
(C) embedding (D) parse grammar only
9. A transformer avoids recurrence by using
(A) attention and positional information (B) only convolutions
(C) only HMMs (D) no token order
10. If 6 of 10 retrieved documents are relevant, precision is
(A) 0.6 (B) 0.4
(C) 1.6 (D) 6.0
11. A parse tree represents
(A) hierarchical syntactic structure (B) document frequency
(C) audio amplitude (D) word embeddings only
12. A dataset can contain bias when its language or labels
(A) underrepresent groups (B) are tokenized
(C) use vectors (D) have punctuation
13. BERT is primarily an
(A) encoder-style bidirectional model (B) decoder-only model
(C) HMM (D) finite-state lexer
14. Recall in retrieval is relevant results divided by
(A) all relevant results (B) retrieved results
(C) all queries (D) true negatives
15. Semantic role labeling identifies predicates and their
(A) arguments and roles (B) characters only
(C) document frequencies (D) attention heads only
16. Autoregressive generation predicts tokens from
(A) previous context (B) future gold tokens
(C) random classes (D) image regions only
17. Lower perplexity generally indicates
(A) better predictive performance (B) worse tokenization al-
ways
(C) higher ambiguity (D) fewer vocabulary items
18. Multi-head attention learns
(A) multiple representation subspaces (B) multiple databases
(C) one token only (D) a single scalar
19. A query with 4 relevant retrieved results and 2 irrelevant results
has precision
(A) 4/6 (B) 2/4
(C) 4/10 (D) 6/4
20. Language is difficult because meaning depends on
(A) context and ambiguity (B) font size
(C) file extension (D) network speed only
21. A language model's log probability is useful because products of
probabilities can
(A) underflow numerically (B) increase vocabulary
(C) remove syntax (D) create labels
22. Self-attention relates tokens within
(A) the same sequence (B) different databases
(C) only audio files (D) a single class label
23. Ambiguity occurs when a sentence has
(A) multiple plausible analyses (B) no tokens
(C) only one word (D) no grammar
24. BLEU evaluates translation using
(A) n-gram overlap (B) entity spans only
(C) word vectors only (D) audio quality only
25. An n-gram model predicts a token from
(A) the previous n-1 tokens (B) the next document only
(C) image pixels (D) a parse tree only
26. The Markov assumption limits dependence to
(A) a fixed history (B) all future tokens
(C) the full corpus always (D) no context
27. Coreference resolution links expressions that refer to the same
(A) entity (B) token count
(C) parse rule (D) audio frame
28. The forward algorithm computes total probability using
(A) dynamic programming (B) greedy stemming
(C) TF-IDF (D) beam width only
29. Cosine similarity measures the angle-based similarity between
(A) vectors (B) parse trees only
(C) documents by length only (D) probabilities only
30. Masked language modeling hides tokens and predicts the
(A) masked tokens (B) next document
(C) POS dictionary (D) audio waveform only
31. Distributional semantics is based on words occurring in similar
(A) contexts (B) fonts
(C) file names (D) audio channels
32. A stop word is often removed because it is
(A) very frequent and less informative (B) always misspelled
(C) a named entity (D) a rare technical term
33. ASR converts speech audio to
(A) text (B) text to images
(C) tokens to audio only (D) labels to vectors
34. The smallest meaningful linguistic unit is a
(A) morpheme (B) document
(C) paragraph (D) parse forest only
35. Constituency parsing groups tokens into
(A) phrases or constituents (B) database tables
(C) attention scores only (D) classes
36. The vector space model represents documents as
(A) vectors of term weights (B) parse graphs only
(C) database schemas (D) class labels
37. Smoothing assigns some probability to
(A) unseen events (B) only frequent words
(C) punctuation only (D) parse roots
38. Word sense disambiguation selects the intended
(A) meaning of a word (B) sentence length
(C) POS tag only (D) document ID
39. Perplexity measures how well a language model
(A) predicts a sequence (B) tags images
(C) retrieves documents only (D) detects entities only
40. Stemming usually removes affixes using
(A) heuristic rules (B) a full world model
(C) audio filters (D) database joins

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1. Tokenization splits text into
(A) tokens (B) class weights
(C) parse probabilities only (D) image regions
2. GPT is primarily a
(A) decoder-only autoregressive model (B) bidirectional encoder only
(C) database engine (D) speech recognizer only
3. An NLP system should protect
(A) private and sensitive text (B) only stop words
(C) only common nouns (D) only punctuation
4. A regular expression is a pattern for matching
(A) text strings (B) neural weights
(C) database rows (D) audio frequencies only
5. Attention lets a decoder focus on
(A) relevant source representations (B) all tokens equally always
(C) only the final token (D) training labels only
6. Bigram probability models depend on
(A) one previous token (B) two previous tokens exactly
(C) the whole paragraph (D) only the label
7. TTS converts text into
(A) speech audio (B) images
(C) parse trees (D) database rows
8. Positional encoding supplies information about
(A) token order (B) word frequency only
(C) document labels (D) audio pitch only
9. In regex, a period commonly matches
(A) a single character (B) a digit only
(C) the end of input only (D) a word boundary only
10. RNNs model sequences using a recurrent
(A) hidden state (B) inverted index
(C) parse node (D) vocabulary hash
11. NLP concerns computational processing of
(A) human language (B) electrical circuits
(C) database pages only (D) image pixels only
12. Seq2seq models map an input sequence to
(A) an output sequence (B) one scalar only
(C) a database row (D) a fixed image
13. A 3-word sentence has how many contiguous bigrams?
(A) 2 (B) 3
(C) 1 (D) 4
14. Precision in retrieval is relevant results divided by
(A) retrieved results (B) all relevant results
(C) all documents (D) false negatives
15. If a term appears in 2 of 10 documents, its document frequency is
(A) 2 (B) 5
(C) 8 (D) 10
16. In parsing, a terminal symbol is typically a
(A) word or token (B) grammar variable
(C) document ID (D) hidden state only
17. One-hot encoding represents each word as
(A) a sparse indicator vector (B) a probability table
(C) a parse tree (D) a dense audio signal
18. The cosine similarity of identical nonzero vectors is
(A) 1 (B) 0
(C) -1 (D) undefined always
19. An HMM contains hidden states and
(A) observations (B) attention layers
(C) database indexes (D) word vectors only
20. NER identifies spans such as people, places, and
(A) organizations (B) grammars
(C) vectors (D) n-grams only
21. In an HMM, POS tags can be hidden states and words can be
(A) observations (B) transitions
(C) rewards (D) documents
22. POS tagging assigns each token a
(A) syntactic category (B) vector norm
(C) document ID (D) sentiment score only
23. A beam search keeps a fixed number of
(A) best partial hypotheses (B) training documents
(C) POS tags (D) attention matrices only
24. ROUGE is commonly used for
(A) summarization overlap (B) POS tagging
(C) speech synthesis (D) database retrieval only
25. Machine translation converts text between
(A) languages (B) formats only
(C) databases (D) audio channels only
26. IDF downweights terms that occur in
(A) many documents (B) one document only
(C) no documents (D) one sentence only
27. Bag-of-words ignores
(A) word order (B) word identity
(C) document membership (D) term counts
28. The log of a probability between 0 and 1 is
(A) nonpositive (B) always positive
(C) always zero (D) undefined
29. Character n-grams can help handle
(A) unknown word forms (B) database transactions
(C) image rotations (D) reward discounting
30. Lemmatization reduces word forms to a
(A) dictionary base form (B) random stem
(C) character code (D) sentence label
31. In scaled dot-product attention, scores use queries and
(A) keys (B) labels
(C) stems (D) documents only
32. An FSA recognizes regular languages using states and
(A) transitions (B) gradients
(C) attention heads (D) parse probabilities
33. Classification accuracy is correct predictions divided by
(A) all predictions (B) predicted positives only
(C) actual positives only (D) retrieved documents
34. A noun phrase commonly contains a
(A) noun head (B) verb root only
(C) transition matrix (D) discount factor
35. Dependency parsing represents head-dependent
(A) relations (B) pixel neighborhoods
(C) document counts (D) reward paths
36. TF measures a term's frequency in a
(A) document (B) corpus only
(C) query class (D) parse tree
37. Word embeddings represent words as
(A) dense vectors (B) binary labels only
(C) regular expressions (D) parse rules only
38. An inverted index maps terms to lists of
(A) documents or positions (B) POS tags only
(C) embedding dimensions (D) audio frames
39. A free morpheme can
(A) stand alone as a word (B) never occur alone
(C) only be a suffix (D) only be punctuation
40. A context-free grammar uses productions with a
(A) single nonterminal on the left (B) probability only
(C) vector on the left (D) regular expression only